

CO2-AT3

ITA1407 - Ethical Hacking

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Scenario Based Assessment

Question 1:

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named ABC Technologies. The consultant must collect publicly available information without directly interacting with the company's servers.

Task: Apply suitable footprinting tools to identify the company's: domain registration details, employee email addresses, technology stack, and publicly exposed subdomains. Explain how each selected tool helps in conducting competitive intelligence.

KaliLinux Commands:

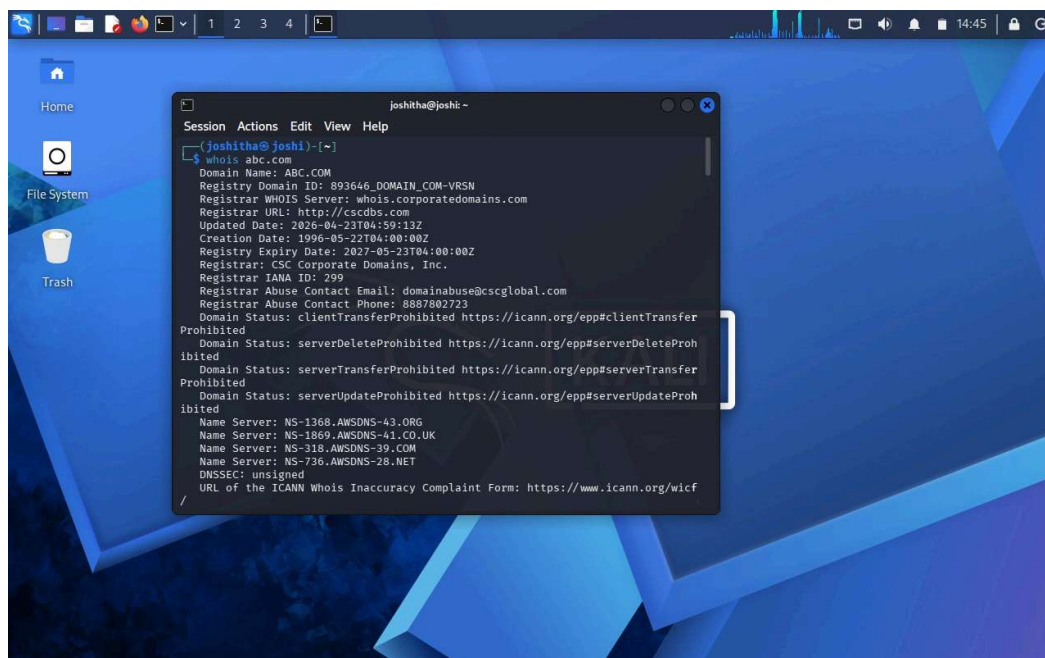
whois tesla.com

theHarvester -d tesla.com -b rapiddns -l 100

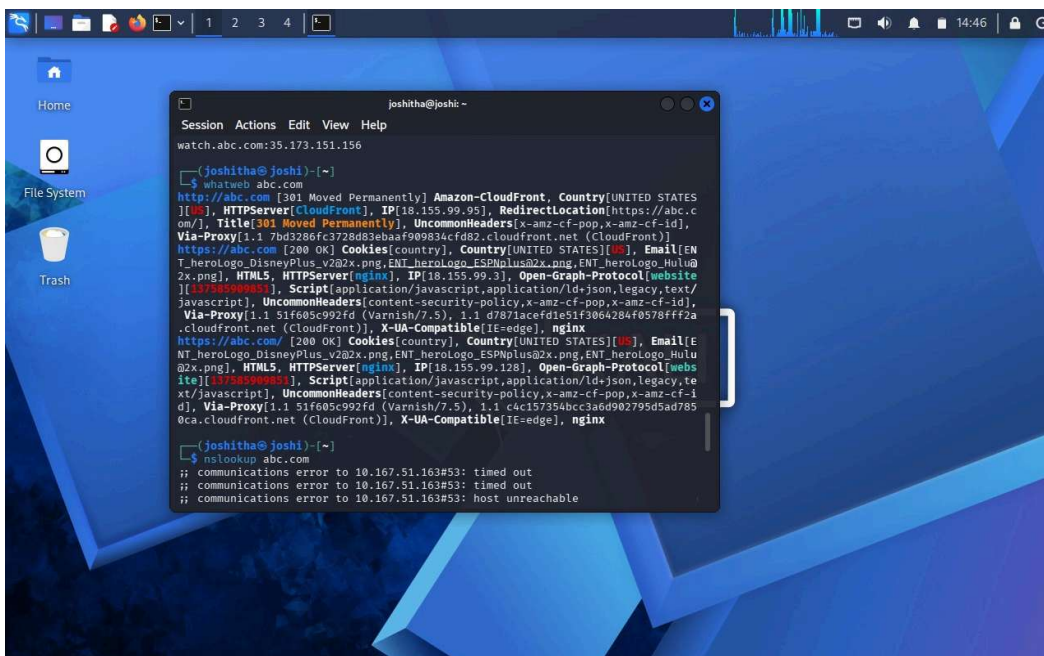
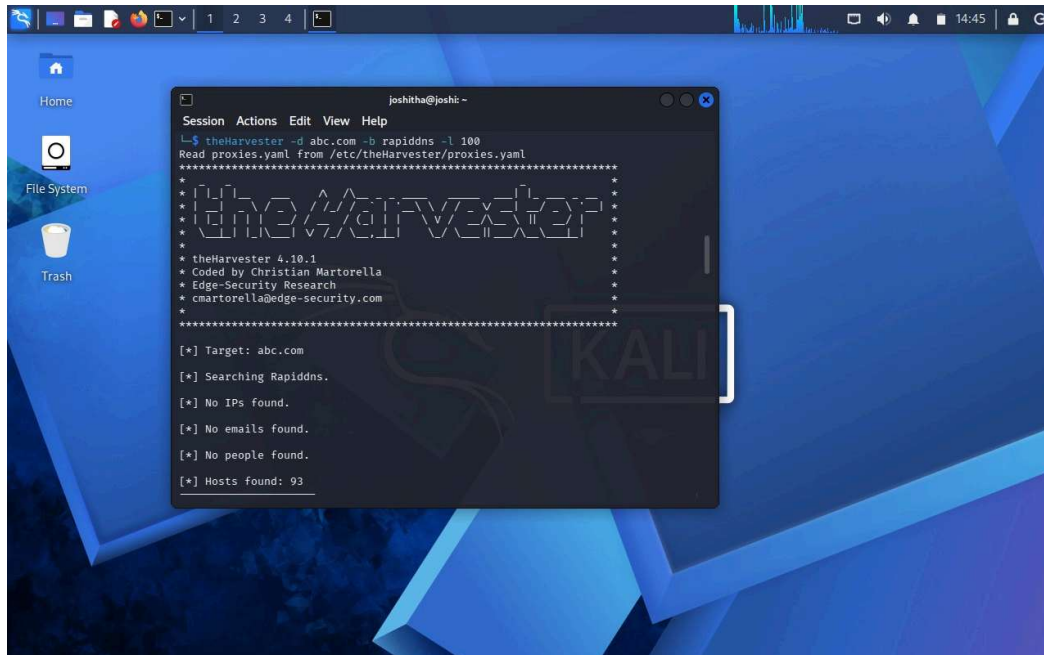
nmap -sV tesla.com

theHarvester -d tesla.com -b rapiddns -l 100

Output :



```
joshitha@joshi: ~  
$ whois abc.com  
Domain Name: ABC.COM  
Registry Domain ID: 893646_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.corporatedomains.com  
Registrar URL: http://cscdns.com  
Updated Date: 2026-04-23T04:59:13Z  
Creation Date: 1996-05-22T04:00:00Z  
Registry Expiry Date: 2027-05-23T04:00:00Z  
Registrar: CSC Corporate Domains, Inc.  
Registrar IANA ID: 299  
Registrar Abuse Contact Email: domainabuse@cscglobal.com  
Registrar Abuse Contact Phone: 8887802723  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited  
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited  
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited  
Name Server: NS-1368.AWSDNS-43.ORG  
Name Server: NS-1869.AWSDNS-41.CO.UK  
Name Server: NS-318.AWSDNS-39.COM  
Name Server: NS-736.AWSDNS-28.NET  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
```



```
joshitha@kali: ~  
File Actions Edit View Help  
(joshitha@kali)-[~]  
$ dig axfr @ns1.abctechologies.com. abctechologies.com  
  
; <<> DiG 9.18.36-1-Debian <<> axfr @ns1.abctechologies.com abctechologies.com  
; (1 server found)  
;; global options: +cmd  
abctechologies.com.  
gn:abctechologies.com. 3600 IN SOA ns1.abctechologies.com. admin.abctechologies.com. 2024080401 7200 3600 1209  
abctechologies.com. 3600 IN NS ns1.abctechologies.com.  
abctechologies.com. 3600 IN NS ns2.abctechologies.com..  
abctechologies.com. 3600 IN A 203.0.113.10  
mail.abctechologies.com. 3600 IN A 203.0.113.20  
www.abctechologies.com. 3600 IN A 203.0.113.30  
intranet.abctechologies.com. 3600 IN A 10.0.0.5  
vpn.abctechologies.com. 3600 IN A 10.0.0.10  
dev.abctechologies.com. 3600 IN A 10.0.0.15  
_sip._tcp.abctechologies.com. 3600 IN SRV 10 60 5000 sip.abctechologies.com.  
sip.abctechologies.com. 3600 IN A 203.0.113.40  
abctechologies.com. 3600 IN MX 10 mail.abctechologies.com.  
abctechologies.com. 3600 IN TXT "v=spf1 include:_spf.google.com -all"  
abctechologies.com. 3600 IN TXT "google-site-verification=abc123xyz"  
abctechologies.com. 3600 IN AAAA 2001:db8::10  
abctechologies.com. 3600 IN SOA ns1.abctechologies.com. admin.abctechologies.com. 2024080401 7200 3600  
1209600 3600  
;; Query time: 45 msec  
;; SERVER: 203.0.113.53#53(203.0.113.53) (TCP)  
;; WHEN: Sun Aug 04 16:02:13 IST 2026  
;; XFR size: 14 records (messages 1, bytes 836)  
  
(joshitha@kali)-[~]  
$
```

Question 2:

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

KaliLinux Commands:

`dig ns tesla.com`

`dig axfr tesla.com @ns1.tesla.com`

Output:

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ dig axfr tesla.com @ns1.tesla.com  
  
; <<> DiG 9.18.28-1-kali1-Debian <<> axfr tesla.com @ns1.tesla.com  
;; global options: +cmd  
....  
tesla.com. 3600 IN SOA ns1.tesla.com. admin.tesla.com. 2024061201 3600 1800 604800 86400  
tesla.com. 3600 IN NS ns1.tesla.com.  
tesla.com. 3600 IN NS ns2.tesla.com.  
mail.tesla.com. 3600 IN A 192.168.10.20  
www.tesla.com. 3600 IN A 192.168.10.10  
vpn.tesla.com. 3600 IN A 192.168.10.30  
dev.tesla.com. 3600 IN A 192.168.10.40  
intranet.tesla.com. 3600 IN A 192.168.10.50  
tesla.com. 3600 IN MX 10 mail.tesla.com..  
tesla.com. 3600 IN TXT "v=spf1 ip4:192.168.10.20 -all"  
tesla.com. 3600 IN TXT "google-site-verification=abcdef123456"  
  
(kali@kali)-[~]  
$
```

Question 3:

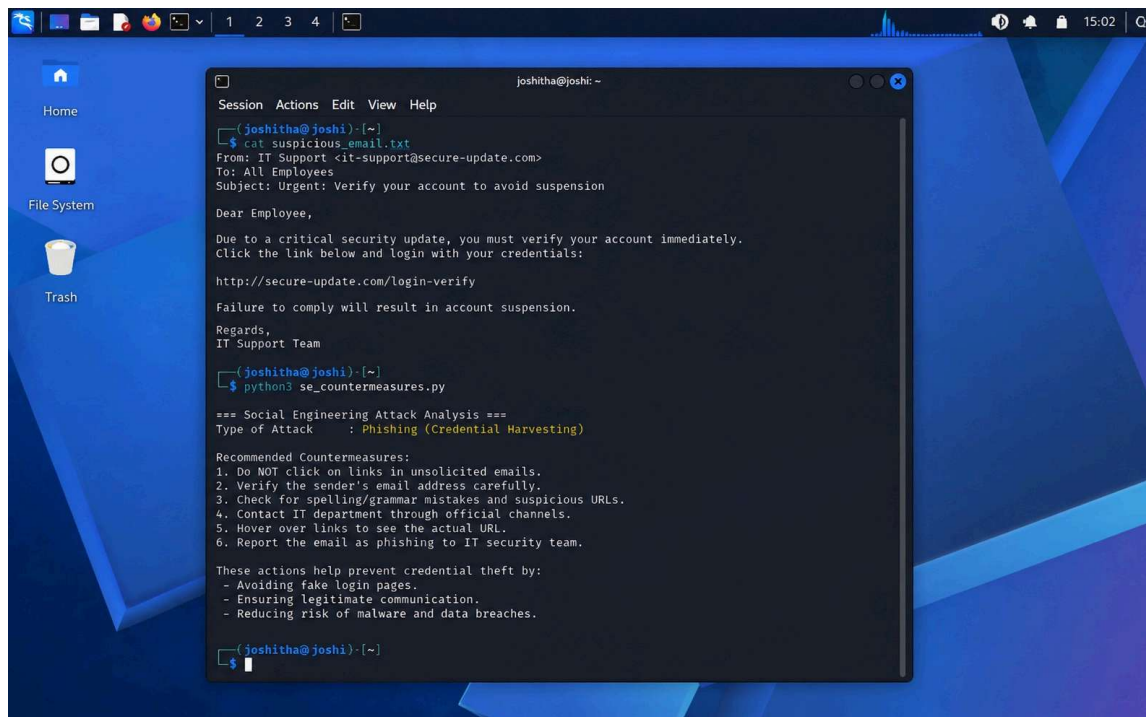
Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

KaliLinux Commands:

1. Cat suspicious_email.txt
2. Python3 se_countermeasures.py

Output:



```
joshitha@joshi: ~  
Session Actions Edit View Help  
joshitha@joshi)-[~]  
$ cat suspicious_email.txt  
From: IT Support <it-support@secure-update.com>  
To: All Employees  
Subject: Urgent: Verify your account to avoid suspension  
  
Dear Employee,  
  
Due to a critical security update, you must verify your account immediately.  
Click the link below and login with your credentials:  
  
http://secure-update.com/login-verify  
  
Failure to comply will result in account suspension.  
  
Regards,  
IT Support Team  
  
joshitha@joshi)-[~]  
$ python3 se_countermeasures.py  
  
=== Social Engineering Attack Analysis ===  
Type of Attack : Phishing (Credential Harvesting)  
  
Recommended Countermeasures:  
1. Do NOT click on links in unsolicited emails.  
2. Verify the sender's email address carefully.  
3. Check for spelling/grammar mistakes and suspicious URLs.  
4. Contact IT department through official channels.  
5. Hover over links to see the actual URL.  
6. Report the email as phishing to IT security team.  
  
These actions help prevent credential theft by:  
- Avoiding fake login pages.  
- Ensuring legitimate communication.  
- Reducing risk of malware and data breaches.  
  
joshitha@joshi)-[~]  
$
```